Julià Delos Ayllón

Software Engineer · C/C++, Linux, Embedded Systems, WebAssembly

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Technical skills

Languages: C, C++, Python, TypeScript/JavaScript, Java, MATLAB, Solidity

Build and tooling: Git with feature branches and pull requests, Git LFS, CMake (built Blender, OpenFOAM, Qucs and a Qucs WebAssembly port from source), Make, GitHub Actions, Docker

Platforms: Linux (daily development on Manjaro), embedded Linux, WebAssembly, dSPACE real-time hardware (MATLAB/Simulink benchmarking)

Graphics: WebGL and GLSL vertex and fragment shaders through Three.js and React Three Fiber; no production OpenGL, Vulkan or Metal work yet

Projects

Switched-capacitor converter impedance tools Mar 2015 – Oct 2026

- Ported my MATLAB converter models to C++ (built on SymEngine) and WebAssembly, with parity tests against MATLAB.
- Set up GitHub Actions builds that compile a pinned third-party library (SymEngine) from source and test the C++ port against it on every pull request.

Experience

Founder, La Calçotada – Netherlands Mar 2014 – Oct 2026

- Designed and built tested TypeScript/Astro DApps and Solidity contracts, with their integrations, documentation and delivery workflows.
- Built CI that tests a Solidity/TypeScript protocol on every pull request and syncs its verified release artifacts into the DApp repository by pull request.
- Presented the venture in investor conversations during the 2024–2025 fundraising phase.

Founder and CTO, INNER / Innertech.ai – Eindhoven, Netherlands Apr 2022 – Dec 2022

- Developed the venture's patent strategy and competitor analysis to inform positioning and venture decisions.
- Built the venture's financial model, go-to-market scenarios and grant proposals.
- Built the venture's Gatsby, React and Sanity landing site.

Co-Founder and CTO, Ripatrip – Remote (team in Portugal) Sept 2019 – June 2020

- Led two junior developers, defined product scope and software architecture, and delivered the full travel-discovery web application under changing requirements.
- Built the Gatsby front end, Sanity content schemas and the scripts that imported travel data, authoring 85% of commits (677 of 797).

Innovation and Technology Consultant, ICT Group – Eindhoven, Netherlands Sept 2016 – Sept 2018

- Scouted blockchain, data-science and IoT opportunities and translated them into business perspectives for ICT Group's business units.
- Guided early data-science projects, then convinced senior leadership to create a dedicated data-science team and data lake.

Research Scientist, Philips Research – Eindhoven, Netherlands June 2011 – Dec 2015

- Defined and validated new circuit concepts for miniaturized LED drivers through models, simulations and prototypes, resulting in publications and invention disclosures.
- First-named inventor on eight EPO patent records (six granted) in integrated power electronics; contributed to the internal miniaturization roadmap.
- Framed innovation concepts with Lean Startup and the Business Model Canvas to tie proposed circuits to business needs.

Hardware Engineer, Lear Corporation – Valls, Spain Mar 2010 – May 2011

- Delivered a working first-iteration 60 kW racing-car inverter, owning its hardware from design and test setup to production.
- Held the inverter's volume target when others called it infeasible, aligning suppliers and technical contributors around it.
- Designed an in-house, very low-profile tangential current sensor at one-sixth of the cost of the off-the-shelf sensor it replaced, with adjustable sensitivity and very high accuracy.

Embedded Systems Engineer, Printer Firmware, HP (via Capgemini) – Barcelona Sept 2006 – Sept 2007

- Developed and maintained HP Large-Format Printers firmware in C and C++ on Linux, shipping official releases and patches.
- Delivered bug fixes in a C/C++ printer-firmware codebase so large that a full build took six hours on a 2007 HP workstation, covering its rendering engine, state machine and peripherals.
- Shipped embedded-Linux security releases with updated third-party libraries.
- Trained the Bangalore engineers taking over the department's firmware during a two-to-three-week transfer visit.

Education

Eindhoven University of Technology, PhD in Electrical Engineering 2016

Technical University of Catalonia, MSc in Electronics Engineering 2010

Polytech Lille, Degree in Industrial Software Development 2006

Technical University of Catalonia, BSc in Electrical Engineering 2005

Languages

Catalan and Spanish (native) · English (fluent) · French and Portuguese (conversational) · Dutch (basic)